SectraTEK

AI FOR CYBERSECURITY AND DIGITAL FORENSICS · PRACTICAL 2

# Can you trust what you hear?



Screening audio evidence for tampering with unsupervised machine learning.

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YOUR INSTRUCTOR

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## Mobile Threat Research Lead

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IN ONE LINE

I study how phones get spied on and tampered with, and build AI tools that help examiners find the evidence.

**Mobile threat research**
Spyware, stalkerware and device tampering.

**AI for investigators**
Machine learning that speeds up the search.

**Teaching**
AI for Cybersecurity and Digital Forensics.

TODAY'S STORY

# From a suspicious recording to an anomaly detector

1

## Fundamentals

How sound and tampering work

2

## Before AI

How examiners checked

3

## The problem

The question we must answer

4

## Manual solutions

What was tried over the years

5

## The gap

Where the old ways fall short

6

## How AI helps

Machine learning and LLMs

7

## Our approach

Unsupervised learning, step by step

8

## Practical

Open the notebook in Colab

Practical 2 uses one notebook: audio_forensic_using_ai.ipynb.

1 FUNDAMENTALS

# A recording can decide a case

### Phone calls

Calls and voice notes with threats, agreements and admissions, saved on a phone.

### Interviews

Statements recorded by police, lawyers and employers.

### Body-cam audio

The sound track of what happened at a scene.

### Private recordings

Made by a witness or a whistleblower, often in secret.

If someone edits a recording, the story it tells changes. A court needs to know it has not been altered.

1 FUNDAMENTALS

# To a computer, sound is a list of numbers



Each bar is one measurement of the sound, called a sample. Real audio has thousands per second.

## Samples

A microphone measures the air pressure many times a second. Each measurement is one number.

## Sample rate

How many samples per second. 48,000 (48 kHz) is common for video; 44,100 (44.1 kHz) is the CD standard.

## Duration

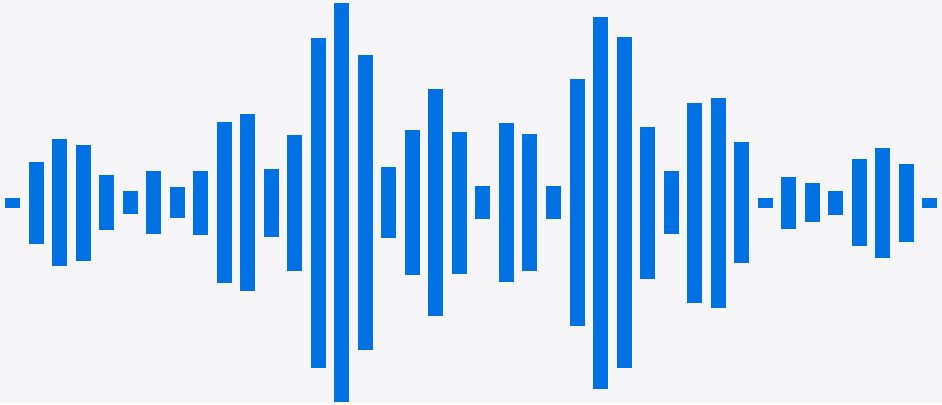
Samples divided by sample rate. 2,880,000 samples at 48 kHz is exactly 60 seconds.

1 FUNDAMENTALS

# Two ways to look at the same sound

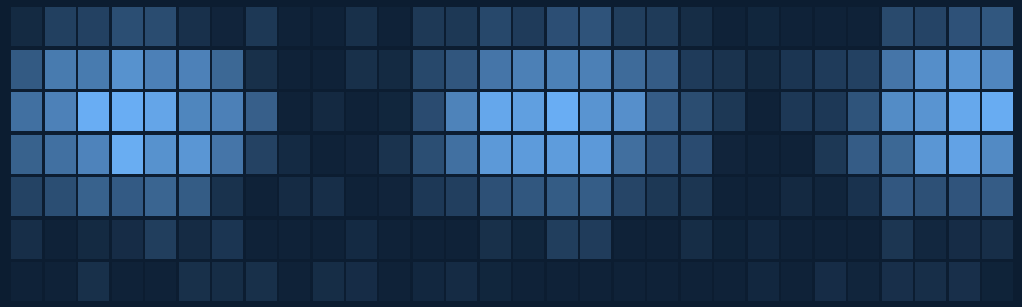
**WAVEFORM**

loudness over time



**SPECTROGRAM**

pitch (frequency) over time



**WHAT AN EDIT LOOKS LIKE**

A clean cut can be silent to the ear, yet it often leaves a sudden break in one of these pictures: a jump in the waveform or a vertical seam in the spectrogram. Reading them is a core forensic skill.

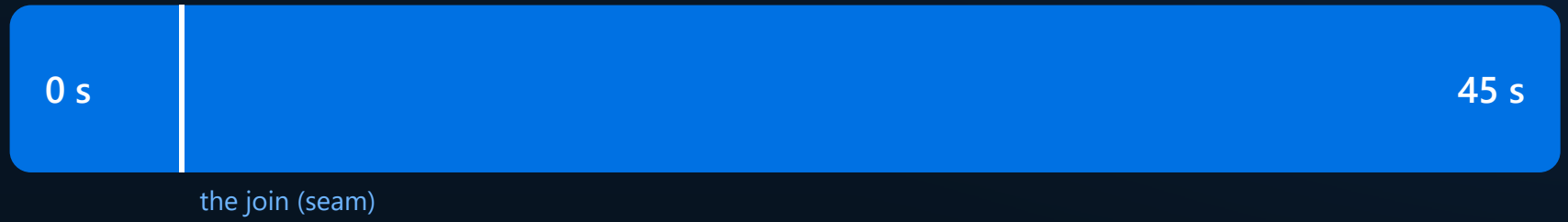
1 FUNDAMENTALS

# The anatomy of a splice

ORIGINAL RECORDING




AFTER THE CUT, THE ENDS JOINED




**The duration drops from 60 s to 45 s.** The listener may hear nothing, but the file is now a different length, which a computer can measure.

1 FUNDAMENTALS

# How a recording gets tampered with

### Cut

Remove words or sentences, then join the two ends.

### Insert

Add words or sounds taken from another recording.

### Re-save

Convert, compress or re-sample the file. This can wipe traces of an edit.

### Synthesize

Generate a fake voice with AI, often called a deepfake.

**TODAY'S FOCUS**

Cuts and re-saving often change a file's length or sample rate. Those are the traces our model will look for.

1 FUNDAMENTALS

# The new threat: voices made by AI

### Cloned in seconds

A few seconds of someone's voice is now enough to make a convincing fake of them saying anything.

### Vishing at scale

Voice phishing scam calls that use a familiar voice, a boss or a relative, to pressure a victim.

### Fake audio as evidence

A recording offered in a case may be genuine, edited, or entirely synthetic. Someone has to tell.

### Why this makes audio forensics critical

As cloned voices spread into fraud and into courtrooms, every important recording needs checking. Detecting edits and synthetic speech is becoming a routine part of digital forensics and legal cases.

2 BEFORE AI

# How examiners checked a recording

### Listen

Careful listening for clicks, breaths cut short, or background noise that suddenly changes.

### Look

Waveforms and spectrograms turn sound into a picture that can be searched for breaks.

### Check the file

Metadata, file structure, hashes, and the recording device's usual format.

### Mains hum (ENF)

The faint hum from the power grid, compared with the grid's own records.

All four need a trained expert, take hours per file, and two experts can disagree.

3 THE PROBLEM

# Has this recording been changed since it was made?

### Edits can be inaudible

A clean cut between words can be impossible to hear.

### There is no original

We rarely have the first, untouched copy to compare.

### There are no labels

We almost never have examples of tampered audio to learn from.

The last point changes everything: we cannot train a model with an answer key we do not have.

4 MANUAL SOLUTIONS

# What examiners built over the years

## 1974

### Watergate tape

An expert panel examined an 18.5-minute gap in a White House tape.

## 1996

### Handling rules

AES27: a recommended practice for handling recorded audio evidence.

## 2005

### Mains hum (ENF)

Using the power-grid hum to date and check recordings.

## 2010s

### Lab guides

Written methods for audio authentication, used by labs today.

## 2019

### Cloned voices

Reported fraud cases using AI-cloned voices raise the stakes.

**THE PATTERN**

Each method needs an expert, time, and often the original recording or the grid's records. Meanwhile, editing tools keep getting better.

5 THE GAP

# Where the old ways fall short

Experts are slow.

Edits keep getting cleaner.

1. **No labels.** We rarely have tampered examples to learn from.
2. **Subtle edits.** Modern editors leave few traces you can hear or see.
3. **Slow and subjective.** Each file can take hours, and experts may disagree.
4. **It does not scale.** One case can involve hundreds of recordings.

6 HOW AI HELPS

# Learn what normal sounds like

## Machine learning

**POINTS TO THE ODD ONES OUT**

- Learns from genuine recordings, with no labels
- Flags the files that do not fit
- Gives every file an anomaly score
- Screens hundreds of files in seconds

## Large language models (LLMs)

**HELPS THE EXAMINER**

- Explain findings in plain language
- Help draft examiner notes
- Help write and fix analysis code
- Cannot tell by listening whether audio is real

AI tells us where to listen first. The examiner still checks, explains and decides.

MACHINE LEARNING IN ONE PICTURE

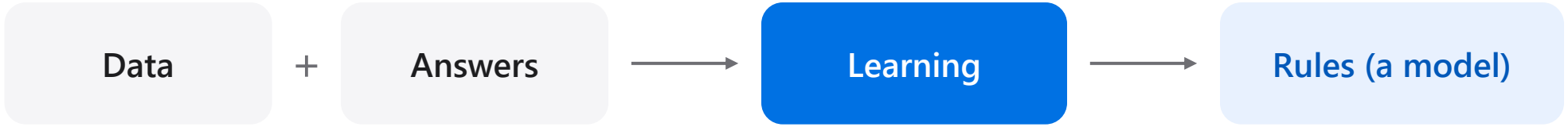
# Writing the rules, or learning them

TRADITIONAL PROGRAMMING




A person writes every rule by hand.

MACHINE LEARNING




The computer works out the rules from examples.

**IN OUR PRACTICAL**

That bottom row is supervised learning. Today we remove the answers: the model gets 614 genuine recordings and must work out what normal looks like on its own.

TWO WAYS A MACHINE CAN LEARN

# Supervised and unsupervised learning

## Supervised learning

### Learning with an answer key

Every training example comes with the right answer, called a label. The model learns to predict the label for new data.

EXAMPLES

- Spam or not spam
- Cover or stego image (Practical 1)
- Malware or clean file

**Algorithms:** Decision Tree, Random Forest, Logistic Regression, Support Vector Machine, k-Nearest Neighbours, Neural Network

TODAY

## Unsupervised learning

### Learning without an answer key

No labels. The model studies the data on its own to find groups, or to spot the odd one out.

EXAMPLES

- Unusual logins at 3 am
- Grouping similar files
- Suspicious audio recordings (Practical 2)

**Algorithms:** k-Means, Isolation Forest, DBSCAN, and PCA for simplifying data

7 OUR APPROACH

# How an unsupervised model learns, in five steps

1

## Collect

Genuine recordings only: 614 of them, with no labels.

2

## Measure

Describe each file with numbers: duration and sample rate.

3

## Learn

The model learns which combinations are common.

4

## Score

Every file gets an anomaly score. Unusual files are flagged.

5

## Decide

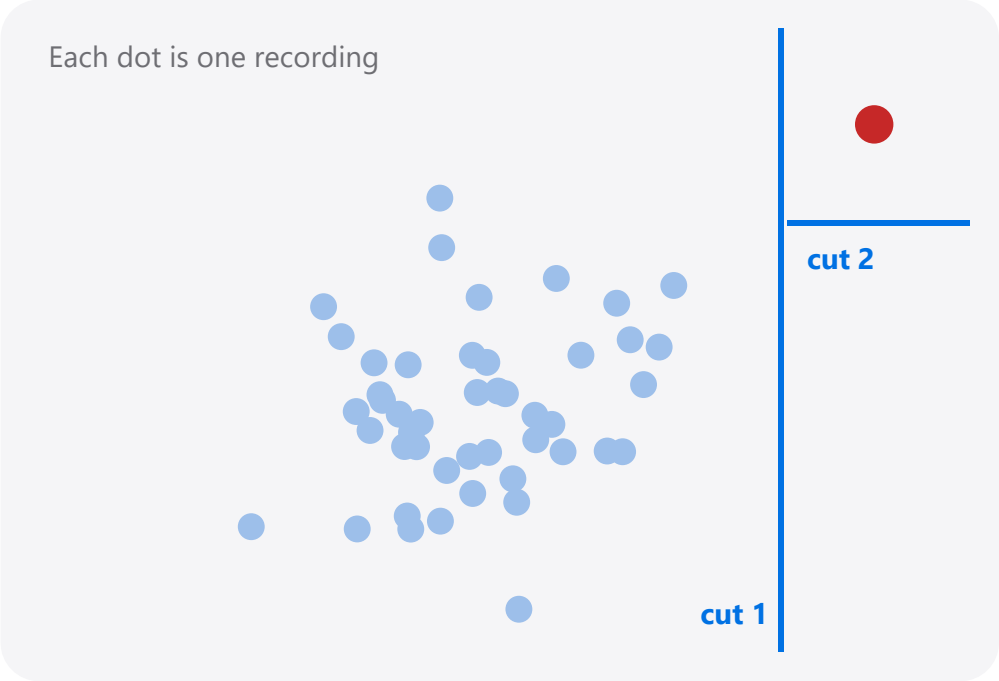
The examiner sets how strict to be and checks the flagged files.

THINK OF A SECURITY GUARD

Nobody told the guard what a thief looks like. After seeing thousands of ordinary visitors, they simply notice the one who does not fit.

7 OUR APPROACH

# Our model: an Isolation Forest




Two random cuts are enough to fence off the odd recording (red). The crowded ones need many more cuts.

## Random cuts

The model draws random lines through the data, again and again.

## Odd points fall out fast

An unusual file is isolated after a few cuts. A normal one needs many.

## 100 trees, one score

Averaged over 100 random trees, the number of cuts becomes an anomaly score.

7 OUR APPROACH

# What the model learns from

**614**

**genuine recordings**

No tampered files, no labels

**2**

**features**

Duration and sample rate

**474**

**at 48 kHz**

Most are about 60 seconds long

**140**

**at 44.1 kHz**

A smaller group, including all the 64 s files

Values.csv: 27 columns → Keep 2 features → Isolation Forest → Flag + score

**Why only two features?** They can be read from any audio file in a fraction of a second, so exactly the same measurements work on new evidence.

7 OUR APPROACH

# Rare is not the same as tampered

**140 of 140**

**genuine 44.1 kHz recordings flagged**

Every file in the minority group, though none was edited.

**183 of 614**

**genuine files flagged in total**

With the default setting, almost 1 in 3 genuine files is flagged.

The model flags what is unusual, not what is fake. A flag means "look closer", never "this was edited".

7 OUR APPROACH

# Testing the model with edits we made ourselves

| | | |
|---|---|---|
| Original | 60.0 s · 48 kHz | NORMAL |
| Spliced: 15 seconds cut out | 45.0 s · 48 kHz | ANOMALOUS |
| Re-sampled to 44.1 kHz | 60.0 s · 44.1 kHz | ANOMALOUS |
| A car horn from the internet | 5.9 s · 48 kHz | ANOMALOUS |

The car horn is flagged correctly, because it is nothing like 60 seconds of speech. That still does not make it tampered.

7 OUR APPROACH

# How strict should the model be?

| SETTING | GENUINE FILES FLAGGED | SPLICED COPY | RE-SAMPLED COPY |
|---|---|---|---|
| auto | 183 of 614 | caught | caught |
| 0.10 | 39 of 614 | caught | missed |
| 0.05 | 31 of 614 | caught | missed |
| 0.01 | 7 of 614 | missed | missed |

STRICT

more false alarms, fewer misses



fewer false alarms, more misses

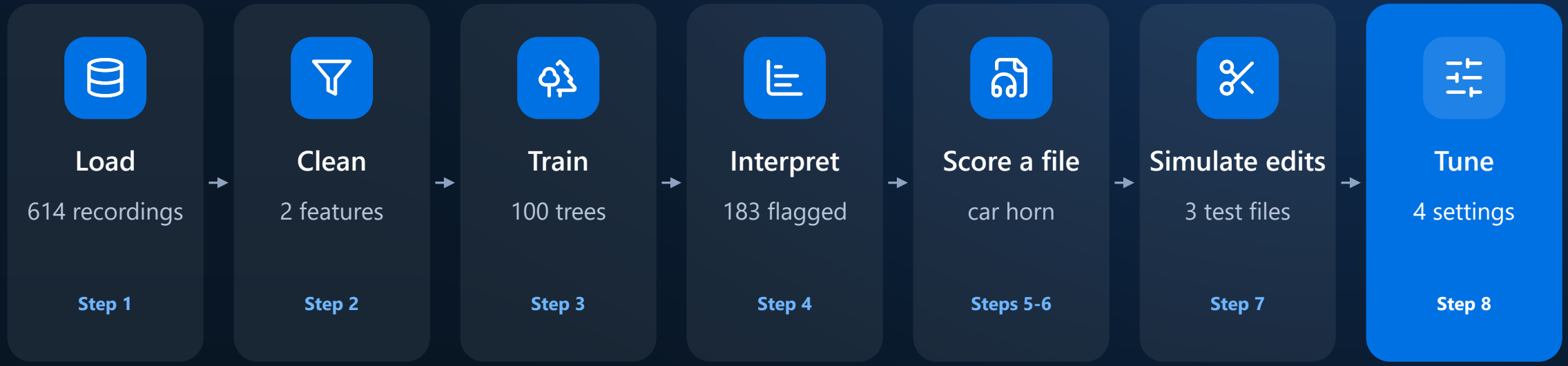
LENIENT

THE TRADE-OFF

There is no perfect setting. It depends on which mistake costs more in your case.

7 OUR APPROACH

# The whole pipeline, mapped to the notebook




Every box is a step you will run. Each step in the notebook starts with its Purpose, Input and Output.

8 PRACTICAL

# Let's open the notebook.

THE NOTEBOOK

## audio_forensic_using_ai.ipynb

Train an Isolation Forest on 614 genuine recordings, score a suspicious file, test it with edits we make ourselves, and tune how strict it is.

Open in Colab



1 **Click Open in Colab**
Nothing to install on your computer.

2 **Runtime, then Run all**
Or press Shift + Enter, one cell at a time.

3 **Read before you run**
Every step starts with Purpose, Input and Output.

The first cell installs anything missing inside Colab. The notebook downloads the dataset and the test audio for you.

8 PRACTICAL

# What each step does

## audio_forensic_using_ai.ipynb

| | |
|---|---|
| Setup | Install what is missing, inside Colab |
| 1 | Load the 614 genuine recordings |
| 2 | Keep 2 features, then look at the data |
| 3 | Train the Isolation Forest |
| 4 | See how many files were flagged, and which kinds |
| 5 | Download a test file, listen to it, measure it |
| 6 | Predict: normal or anomalous, with a score |
| 7 | Make an original, a cut and a re-sampled copy |
| 8 | Try four strictness settings |
| End | Summary and discussion questions |

8 PRACTICAL

# What you should see, and what to discuss

**183**

genuine files flagged with the default setting

**5.9 s**

the car horn: flagged as anomalous

**3 of 3**

simulated files judged as expected (original normal, both edits flagged)

**0.01**

the setting that misses both edits

DISCUSS

1. What would you do with a file the model flags? And with one it does not flag?
2. Which extra features would make the detector harder to fool?
3. How strict would you be if a missed forgery were very costly?

Numbers from the executed notebook. Your run should match because the random seed is fixed.

SectraTEK

# Rare is not the same as tampered.

Unsupervised learning points to what is unusual. The examiner finds out why.



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